Competition Research of Competitive FSAE Tire Compounds

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Introduction

In the Formula Society of Automotive Engineers (FSAE), numerous factors contribute to success. A critical component is the tire that is run on the car. The tire is the starting point for all vehicle dynamic modeling and calculations. Choosing the most suitable compound can make or break the car's overall performance. To help analyze the best compound, a wide range of teams, tire compounds, sizes, and competition results were examined. These will be analyzed to determine the statistically best tire and why.

Background

In this process, only public information will be collected and analyzed. Tire compounds and sizes will be pulled either from team websites or Instagram posts, and as many teams as possible from the top 10 will have data collected from them. Several key terms will be used to describe the overall effect of tires and their performance, including Tire Life, which refers to the longevity or wear over time. Grip, when talking about the "traction" a tire has on the road, is used loosely and is not an accurate representation of tire grip, as it depends on several factors, not just one coefficient called grip. Tire size refers to both the diameter and the width of the tire. Compound talking about the material makeup of a tire.

Methodology

As mentioned previously, all data for this process will be collected from purely online sources, including websites and social media. There is a small sample group for this study due to a lack of general information available about the tires that teams are using and the limited variety. As it is shown, there are mainly only 2 competitive tire compounds that are used in FSAE. This reduces the need to have a much larger sample size, as there are only 2 possibilities to optimize for. There are several ways to compare the two compounds: Width, Diameter, Grip, Wear, Temperature, and Compound. Using these six different metrics, a theoretical best tire can be chosen.

Analysis and Discussion of Team Data

5 teams in the top 10 have shown which tire they use: Purdue, Texas A&M, Georgia Tech, Alabama, and the University of Wisconsin.

Team Name	Tire Description	Acceleration	Skidpad	Autocross	Endurance
Purdue	Hoosier 16.0 * 7.5-10 LC0 on 7in Wheel	93.9	64.1	94.4	213.7
Texas A and M	Hoosier 16.0 * 7.5 - 10 R20	89.7	65.8	106.6	252.6
Georgia Tech	Hoosier 16.0 * 7.5 - 10 R20	80.6	48.1	122.3	275
Alabama	Hoosier 16.0 * 7.5 - 10 R20	87.3	58	103.3	221.7
University of Wisconsin	Hoosier 16.0 * 6.0 - 10 R25	93.2	58.9	88.1	167.1

First, let's define the two tires shown here, the LC0 and the R20, and outline the key differences between them. Another tire seen is the R25, which is the previous generation of the R20. Therefore, this tire will be included in the R20 data, as the most recent version of a compound would be used if you were to purchase new tires.

FORMULA STUDENT (FSAE)

ITEM NUMBER	TIRE SIZE	TREAD PATTERN	TREAD WIDTH	APPROX. DIA.	APPROX CIRC.	RECOM. RIM WIDTH	MEASURED RIM WIDTH	SECTION WIDTH	COMPOUND
41100	6.0/18.0-10	FSAE-Slick	8LBS	6.0"	18.1"	6-7"	7.0"	8.5"	LCO
43070	16.0 x 6.0-10	FSAE-Slick	7LBS	6.0"	16.3"	6-7"	6.0"	6.9"	R20
43075	16.0 x 7.5-10	FSAE-Slick	8LBS	7.3"	16.2"	6-8"	8.0"	8.4"	LCO, R20
43100	18.0 x 6.0-10	FSAE-Slick	9LBS	6.2"	18.1"	6.5-7"	6.0"	8.1"	R20
43104	18.0 x 7.5-10	FSAE-Slick	10LBS	7.5"	18.3"	7-8"	8.0"	9.5"	R20
43164	20.5 x 7.0-13	FSAE-Slick	11LBS	7.0"	21.0"	5.5-8"	6.0"	8.0"	R20
43167	20.0 x 7.5-13	FSAE-Slick	12LBS	8.0"	20.6"	7-9"	8.0"	9.4"	R20
44070	16.0 x 7.0-10	Circuit Wet	7.5LBS	7.0"	16.2"	6-7"	6.0"	7.6"	W3
44115	18.0 x 6.0-10	FSAE-Wet	9LBS	6.2"	17.8"	6-8"	6.0"	7.8"	W3
44125	19.5 x 6.5-10	Circuit Wet	10LBS	6.2"	19.5"	6-8"	6.0"	8.2"	Wet
44150	20.0 x 7.5-13	Circuit Wet	12LBS	7.4"	20.6"	6-8"	7.0"	8.3"	W3
44185	21.0 x 6.5-13	Circuit Wet	11LBS	6.7"	20.9"	6-8"	6.0"	7.2"	W3

(Figure: Hoosier Compound List (Hoosier Racing Tire 2025))

Hoosier LC0

The Hoosier LC0 tire is the softest in the Hoosier compound lineup. One team that uses this compound is Purdue. The reason for selecting the LC0 tire is the soft compound. It can be properly prepped and warmed up before the event, which can lead to better performance in all shorter events, particularly Acceleration, Skidpad, and potentially autocross if your car is designed to handle the tires. The LC0 is more competitive in these shorter events because of the tire compound, which is “softer”. That softness gives the tire more grip in these short events, so it won't overheat or wear enough to be out of peak performance. Another sign of this is the scoring data: while not fully indicative, given the many other variables that affect car performance, the data still makes sense. Purdue scored first in acceleration, which would be the strongest event for a soft compound tire. Second in skidpad and then fourth in autocross and endurance. This aligns with our understanding of soft tires: as the operating temperature rises over longer events, the tires will no longer be in the peak window.

Hoosier R20

The Hoosier R20 tire is the second softest in the Hoosier compound lineup. The rest of the teams, besides Purdue, chose to use this tire. The R20 is still a soft tire compound according to Hoosier; however, it is significantly more durable than the LC0, which is the softest of the tire compounds provided by Hoosier. The R20 is the best all-around tire based on the data gathered, scoring first in Skidpad, Autocros, and Endurance. The performance across all events except one shows that it is the better compound to use for the events currently in FSAE. Acceleration is 100 points compared to the 475 points from the rest of the events, where the R20 is the better tire.

Tire Size

Tire size is relatively simple compared to compound, as it is a physics-based relationship. The smaller the diameter of the tire, the less moment of inertia that wheel has. The smaller the moment of inertia (MOI), the less energy it takes to spin that wheel. Allowing for faster acceleration and less force through the brakes. The width is slightly more complicated because, as you change it, the contact patch is tied to that value. Teams don't provide exact width data because Hoosier tires have a range of wheel widths they can fit and still function on. This is best analyzed using a mathematical model to determine the optimal wheel width.

Conclusion

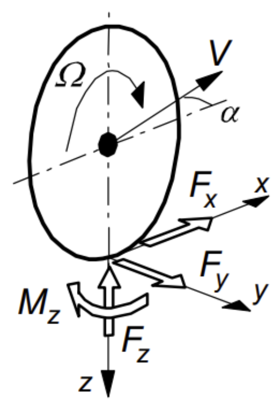
The clear winner in this analysis is the R20. There are three clear reasons for this decision. The first is optimized for most points; 100 points are insignificant compared with the 475 points available from the other events. The second reason is that not everyone using the LC0 teams can still be competitive in acceleration when using the R20 with that tire. The third reason is that if the car is not the strongest in acceleration, optimizing for an event where the vehicle is already behind does not make sense compared to optimizing for the remaining points available.

Models

Several different models can be used to describe different aspects of a tire. There are static and dynamic models that simulate different outcomes at different levels of precision. A popular model still used in industry is Pacejka's tire model. His model describes three forces and provides a mathematical method to predict them. The Lateral Force compared to the slip angle. The longitudinal force compared to the slip ratio, and then the aligning moment compared to the slip angle. These models can be used to analyze different tire compounds side by side to provide a data-driven approximation of performance.

<u>Mathematical explanation of Tire Forces</u>

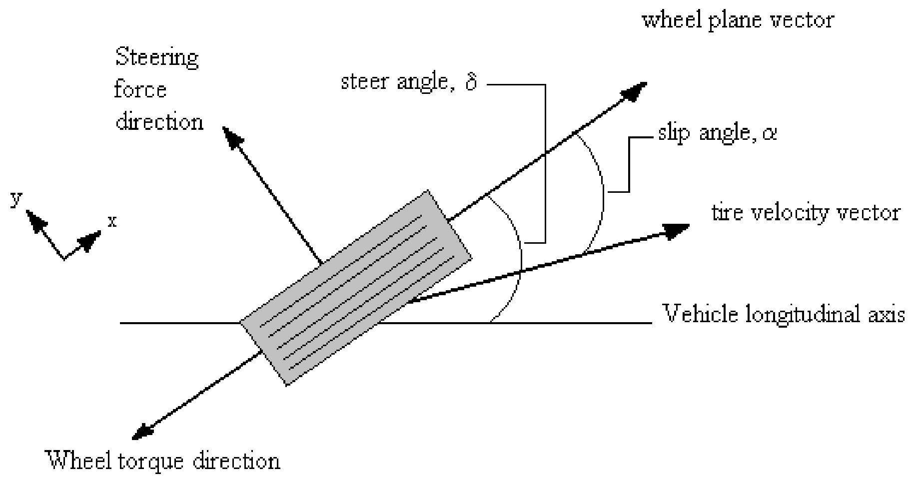
To fully understand the data from a tire model, three characteristics need to be defined and explained. Lateral Force, Longitudinal Force, and the Aligning moment of the tire. These values are important because they can broadly explain what is happening to the tire at any given moment and can be used to derive metrics for further analysis.



(Fig. 2: Tire Force Unit Diagram (Pacejka 2007))

<u>Lateral Slip Angle</u>

The angle between where the tire is pointed and where it is currently going. Slip is generated because of Newton's second law: an object in motion will stay in motion unless acted upon by an external force. In this case, the object in motion is our contact patch. This is the portion of the tire that is in direct contact with the road and points straight. When the wheel is turned, the contact patch is acted on by an opposing force because the sections of the wheel outside the contact patch are moving in the new direction, while the contact patch continues to move in the original direction. This creates an angle between the contact patch and the tire's direction.



(Fig. 3 Lateral Slip Angle Diagram (Suspension Secrets 2017))

Lateral Slip angle is defined as:

$$\tan\alpha = -\frac{V_y}{V_x}$$

(Fig. 4: Lateral Slip Angle Definition (Pacejka 2007))

Longitudinal Slip

Logitudinal Slip occurs when torque is applied to the wheel about its spin axis. This describes the sliding or slipping that can be felt during acceleration and under braking. Longitudinal Slip is defined as:

$$\kappa = -\frac{V_x - r_e\Omega}{V_x} = -\frac{\Omega_o - \Omega}{\Omega_o}$$

(Fig. 5: Longitudinal Slip Angle Definition (Pacejka 2007))

Stiffness

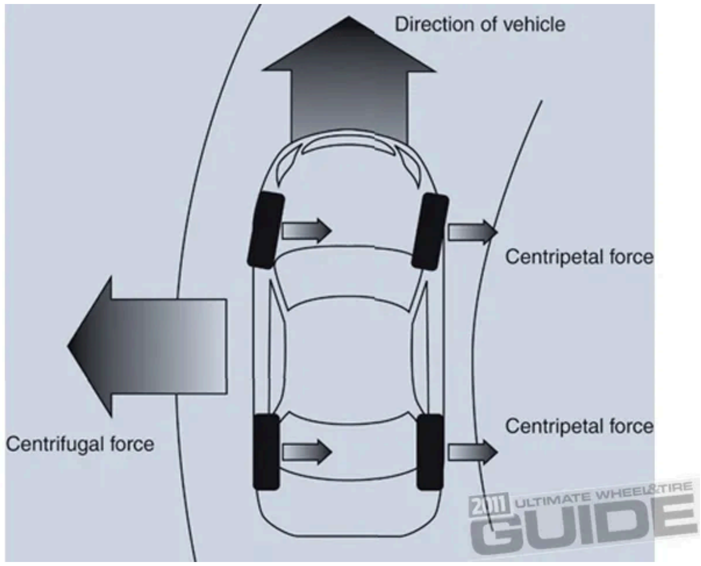
Stiffness is the resistance to slip; both longitudinal (C_{FK}) and lateral/Cornering($C_{F\alpha}$) have a stiffness that, in the equations, acts as a constant to modify the outcome.

$$\begin{aligned} F_x &= C_{F\kappa}\kappa \\ F_y &= C_{F\alpha}\alpha + C_{F\gamma}\gamma \\ M_z &= -C_{M\alpha}\alpha + C_{M\gamma}\gamma \end{aligned}$$

(Fig. 6: Tire Stifness Definition (Pacejka 2007))

Lateral Force

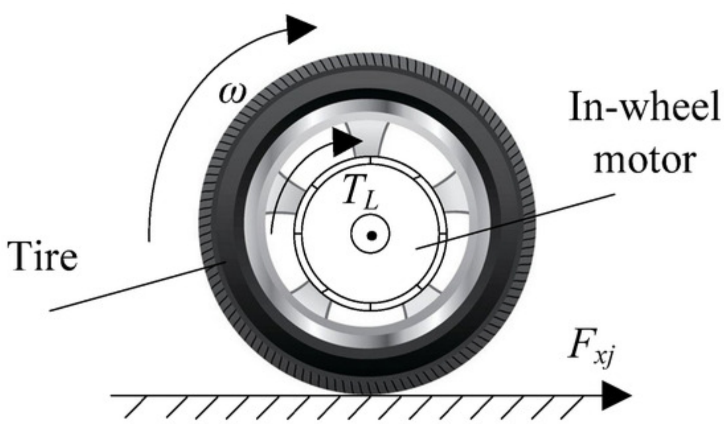
Lateral force is the force that is exerted upon the tire in the Fy direction. This is the force pulling the car into a turn. It counteracts the natural centrifugal force generated by corning. The force is generated by the car's contact patch and is influenced by steering angle and camber, denoted as γ.



(Fig. 6: Lateral Force Diagram (Staff 2011))

Logitudinal Force

Longitudinal Force is the force that is pulling the vehicle in the direction of motion, referenced as F_x. Both acceleration and braking produce changes in force: positive and negative, respectively. Like lateral force, longitudinal force is also generated at the contact patch by the deformation of the wheel.



(Fig. 7: Logitudinal Force Diagram (Wang, Chen, and Lu 2025))

Aligning Moment

Aligning Moment, denoted as M_x, occurs when the cars want the wheel to return to a position where no moment is generated. A change in the steering angle causes the aligning moment. This changes the lateral force, causing the Aligning moment to act. Once the wheel returns to a no-lateral-force condition, the moment stops.

Magic Formula Usage

The magical formula is a simple representation of these forces, creating a basic model of their relative contributions as a function of their respective slip angles. The formulas below show the equation for the lateral force. The values D, C, C1, C2, and mew are all constants that can be determined in different methods from raw data.

$$F_y = D\sin[C\arctan\{B\alpha - E(B\alpha - \arctan(B\alpha))\}]$$

with stiffness factor

$$B = C_{F\alpha}/(CD)$$

peak factor

$$D = \mu F_z \quad (= F_{y,peak})$$

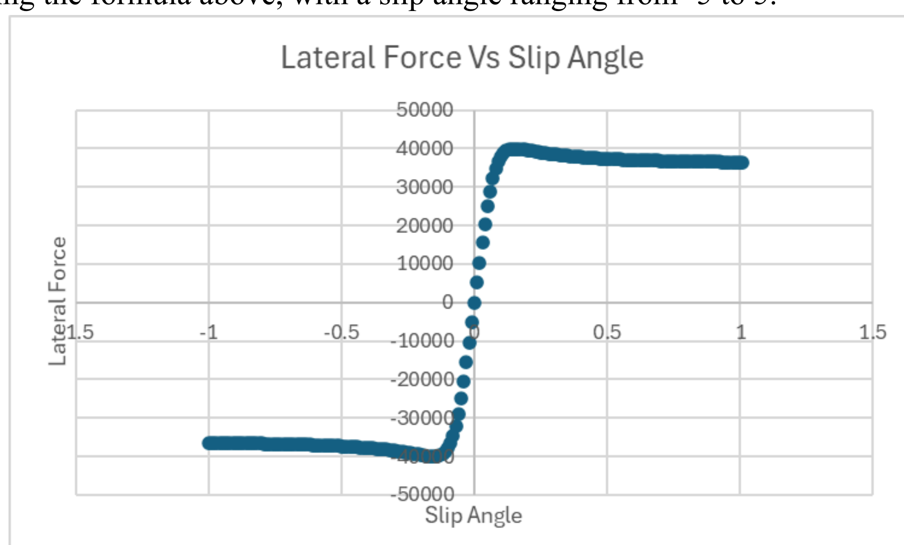
and cornering stiffness

$$C_{F\alpha}\ (= BCD) = c_1 \sin\{2\arctan(F_z/c_2)\}$$

(Fig. 8: Magic Formula Equations (Pacejka 2007))

Usefulness

If used in Excel to graph basic tests, one would begin to see its usefulness. This graph is produced using the formula above, with a slip angle ranging from -5 to 5.



(Fig. 9 Excel Plot of Pacejka Curve)

From the graph above, an engineer can visualize several characteristics of a tire. One that is visible is the Peak Force. This is the maximum value generated. As shown above, that value peaks and then declines, so having a tire with a higher peak or a higher constant value afterward could be advantageous, depending on what you are optimizing for.

Model Methodology

Pacejkas is a model that uses a set of constants, each with a different value, to align a best-fit curve with the actual data, which can then be used to extrapolate results and draw conclusions. Pacejkas is a complex formula, and this paper does not intend to cover its derivation or formulation. However, the goal is to use available Tire Test Consortium(TTC) data, which will then be fitted to a Pacejka curve to demonstrate analytically why it is the best tire choice. Then conclusions will be drawn about things like temperature data, and then the relationships between slip angle and lateral and longitudinal force, as well as the required aligning moment.

<u>Modeling Tool</u>

The modeling tool is developed in C++ and allows TTC data to be visualized and curves to be manually fitted to it. The program just plots the data and separates it by normal force and tire pressure. This allows a simple initial analysis using only the data, and then uses a Pacejka model for a more in-depth analysis.

<u>Data Output</u>

The first step in interpreting a tire model's output is to understand what the graph itself shows. Below is a sample graph labeled with several important zones or characteristics of a lateral force graph.

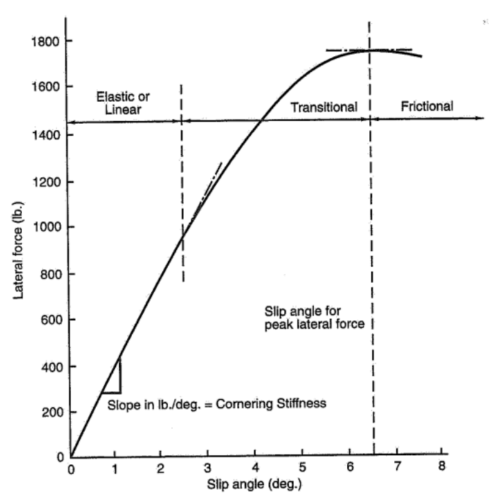


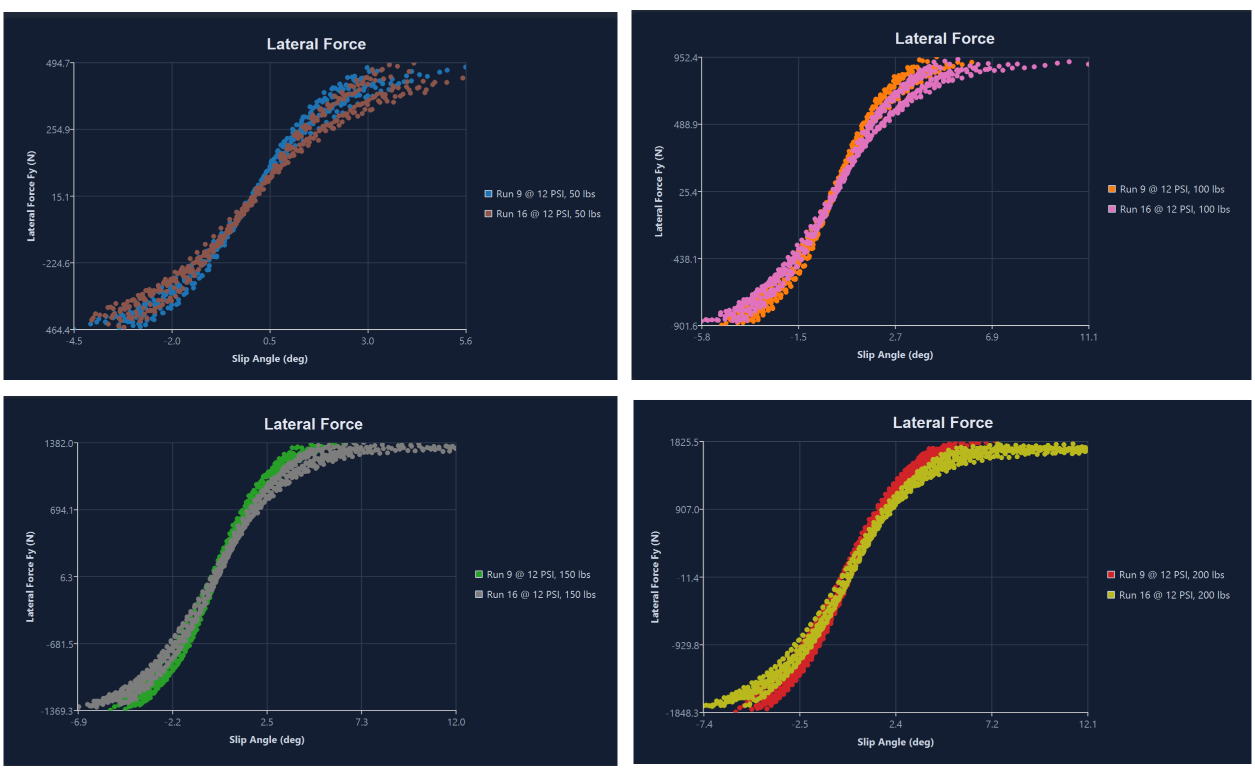
Figure 2.7 Lateral force vs. slip angle for a racing tire.

(Fig. 10 (Milliken and Milliken 1995))

The peak force is labeled at the point where the line's slope is zero. There are three distinct zones, Elastic or linear, which describe when the tire is in full grip with the road. This creates a linear relationship between the slip angle and the lateral force. The transitional zone is when the tire moves from full grip to the beginning of slip. The curve is no longer linear, and it requires a greater slip angle to generate lateral force. The intersection of the transitional and frictional zones is the peak lateral force. The frictional zone, as its name suggests, is where the tire is in friction, sliding instead of generating grip. This is also known as a high-slip condition.

When conducting an in-depth analysis of a full-suspension setup, several other variables are considered; however, one important factor, especially when analyzing Pacejka graphs, is the effect of camber thrust on lateral force and aligning moment. Camber thrust acts like a “preload” for the tires, providing an initial lateral force generated by the setup's geometry. While this can have an initial benefit, not having to start from “ 0 “ lateral force, the peak lateral force can be lowered. As with other aspects of vehicle dynamics, balance is key.

Below are 4 figures, each comparing an LC0 tire with an R20. The raw data points are plotted, providing sufficient evidence to make an initial decision without further analysis. The 3 different graphs are all at 12PSI and are degraded tires; the independent variable is the F_z force or the “ weight of the car. “



(Fig. 11 TTC Tire Force Comparison Graphs)

In these different cases, F_z ranging from 50lb to 200lb, a difference can be seen between the LC0 and the R20. In all diagrams, Run 9 is the R20, and Run 16 is the LC0 tire. In all cases, the R20 produces a higher maximum lateral force than the LC0. Not only does it have a higher peak in the cornering stiffness, but the slope of the linear region is also steeper. So the R20 will generate lateral force more quickly than the LC0, because the tire generates more lateral force and has higher cornering stiffness, and with the previous analysis about optimizing for car type and performance in autocross, skidpad, and endurance. The R20 is the winner.

Conclusion

After doing both an initial results-driven analysis of the two tires, comparing the resulting scores between teams using both the LC0 and the R20. As well as further investigation using the TTC database to back up the initial decision. Given all the available resources, the decision to use the R20 was a clear choice.

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